

TAIS: A Telemetry Assessment and Integrity System for AIS-140 Vehicle Tracking

MCA Capstone Project Proposal — Aryan Bhagat, 25MCA0022, VIT Vellore

Abstract

India's AIS-140 standard mandates GPS/NavIC-based tracking devices on all commercial vehicles, and the government is now building GNSS-based satellite tolling on this same hardware — meaning a vehicle's reported location will soon directly determine how much a vehicle owner is charged. Existing tamper protection only detects physical interference (casing opened, SIM removed); nothing checks whether the telemetry data itself is internally consistent. TAIS is a server-side system that analyzes AIS-140-format telemetry and computes a Trust Score with an explained confidence level, flagging data that is kinematically implausible (spoofed, corrupted, or degraded) for further review. The system is built and evaluated using a simulated telemetry generator — no physical hardware or real-world driving data collection is required to complete the capstone; a real device is explicitly future work.

Problem Statement

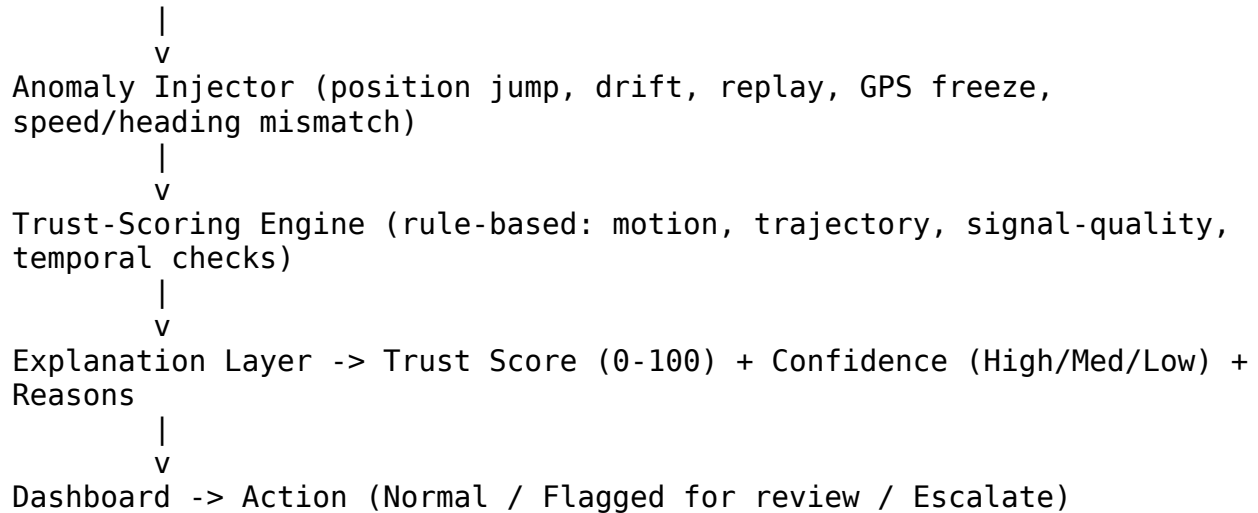
- AIS-140 trackers are mandatory and are the hardware foundation for India's new GNSS tolling rollout.
- Current tamper detection is physical-only — it does not check if the data is trustworthy.
- As tolling becomes fully location-based, inconsistent telemetry becomes a financial risk, not just a compliance one.
- No publicly documented tool assesses AIS-140 telemetry integrity specifically, using only the fields the standard already transmits.

Objectives

1. Build a telemetry simulator producing realistic AIS-140-format data (position, speed, heading, HDOP/PDOP, satellite count, motion-event flags) for varied scenarios (highway, city, degraded-signal).
2. Inject realistic anomalies into simulated data (position jumps, drift, replay, GPS freeze, speed/heading mismatches).
3. Build a rule-based Trust-Scoring Engine that flags kinematically inconsistent telemetry, with a reason and confidence level attached.
4. Build a dashboard demonstrating scoring behavior and a tiered response (Normal / Flagged / Escalate).
5. Evaluate using precision, recall, F1-score, and detection latency against the simulator's own labeled ground truth.

Architecture

Telemetry Simulator (scenario-driven: highway / city / degraded-signal)



Future Work: Telemetry Simulator -> replaceable by a real AIS-140 device feed, without changing anything downstream.

Why server-side, not on-device: the vehicle owner is the party most likely to benefit from faked telemetry, so a check running on their own device can't be trusted. The scoring engine belongs where a transport authority/toll operator would actually receive the data.

Trust Score — Definition

A weighted composite of four sub-scores, computed per telemetry window:

- **Motion Consistency** — is reported speed/acceleration physically achievable?
- **Trajectory Consistency** — is the position sequence a plausible path (no teleports/impossible turns)?
- **Signal Quality** — HDOP/PDOP and satellite count vs. expected thresholds.
- **Temporal Consistency** — timestamps in order, expected intervals, no replay.

Trust Score = $w_1(\text{Motion}) + w_2(\text{Trajectory}) + w_3(\text{Signal Quality}) + w_4(\text{Temporal})$

Initial weights are reasonable defaults based on domain reasoning, refined empirically during evaluation. Every score ships with a Confidence level, so a low score from one ambiguous signal (e.g., poor HDOP — could be a tunnel) is reported as Low Confidence, while multiple independent rule violations together are High Confidence. This is what turns the dashboard into an auditable decision instead of a black-box number.

Threat Model (Scope Boundary)

Assumed trusted: the backend server/storage, and the communication channel's authenticity at the point the server receives it. **Attacker (vehicle owner/operator) can:** control/fabricate what telemetry values are reported before transmission; replay old genuine telemetry as current. **Out of scope:** compromising the backend itself, breaking channel cryptography, physical device tampering (already covered by AIS-140's existing detection).

Existing Approaches vs. This Gap

Approach	Limitation
AIS-140 tamper detection	Physical tampering only, not data consistency
Aviation RAIM (GPS integrity monitoring)	Needs receiver-level satellite data; not adapted to road vehicles or India's mandate
Commercial fleet dashboards	Built for driver-behavior scoring, not telemetry integrity

Modules

1. **Telemetry Simulator** — generates labeled scenario-based AIS-140-format data.
2. **Anomaly Injector** — produces the six anomaly types from genuine simulated logs.
3. **Trust-Scoring Engine** — rule-based checks (ML comparison layer as stretch goal only).
4. **Dashboard** — visualizes scores, reasons, confidence, and tiered actions.
5. **Evaluation Module** — precision/recall/F1/latency scripts against simulator ground truth.

Tools & Technologies

Python (pandas, NumPy) for the simulator and scoring engine; scikit-learn for the optional ML comparison; Flask/Streamlit for the dashboard; GitHub for version control. No hardware required.

Timeline

See `TAIS_Project_Timeline.pdf` in this repository for the full week-by-week schedule.

Real-World Relevance

AIS-140 is a real, enforced Indian mandate, and GNSS-based satellite tolling (announced by MoRTH) is being built on this same hardware. TAIS addresses a genuine current gap — today's tamper detection catches someone opening the device, not telemetry that's become internally implausible — which matters far more once that data decides a bank charge.

Conclusion

TAIS is a fully software-based, achievable capstone: a simulator, a rule-based integrity engine with explainable confidence scoring, and a demo dashboard — solving a real, current, verifiable problem in India's vehicle-tracking infrastructure, with zero hardware or field-data-collection risk.